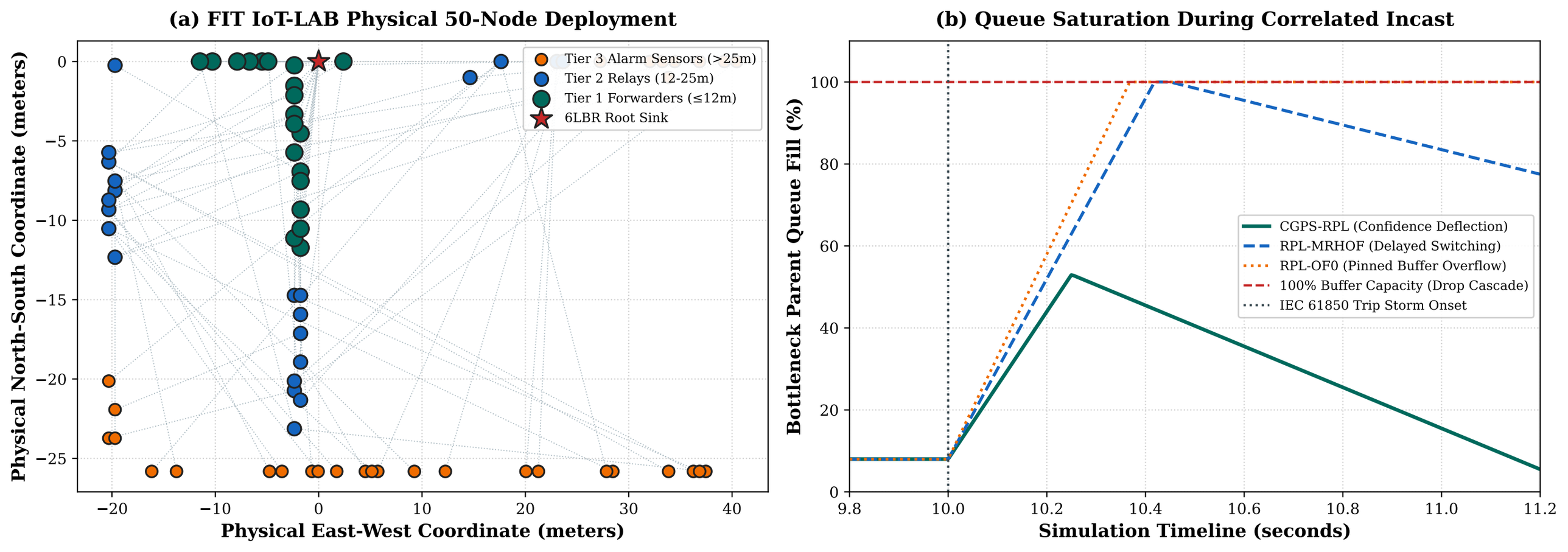
(a) FIT IoT-LAB Physical 50-Node Deployment
Tier 3 Alarm Sensors (>25m)
Tier 2 Relays (12-25m)
Tier 1 Forwarders (≤12m)
6LBR Root Sink
Physical North-South Coordinate (meters)
Physical East-West Coordinate (meters)
0
−5
−10
−15
−20
−25
−20
−10
0
10
20
30
40
(b) Queue Saturation During Correlated Incast
CGPS-RPL (Confidence Deflection)
RPL-MRHOF (Delayed Switching)
RPL-OF0 (Pinned Buffer Overflow)
100% Buffer Capacity (Drop Cascade)
IEC 61850 Trip Storm Onset
Bottleneck Parent Queue Fill (%)
Simulation Timeline (seconds)
0
20
40
60
80
100
9.8
10.0
10.2
10.4
10.6
10.8
11.0
11.2